Alternative Hypothesis: H_1: $\mu_i \neq \mu_j$ for atleast one pair (i, j); $i, j = 1, 2, 3$; $i \neq j$.

That is, there is significant difference in the average reduction in blood pressure in atleast one pair of treatments.

Step 2 : Data

Medication	10	12	9	15	13
Exercise	6	8	3	0	2
Diet	5	9	12	8	4

Step 3 : Level of significance $\alpha = 0.05$

Step 4 : Test statistic

$$F_0 = \frac{MST}{MSE}$$

Step 5 : Calculation of Test statistic

						Total	**Square**
Medication	10	12	9	15	13	59	3481
Exercise	6	8	3	0	2	19	361
Diet	5	9	12	8	4	38	1444
						G = 116	5286

Individual squares

Medication	100	144	81	225	169
Exercise	36	64	9	0	4
Diet	25	81	144	64	16

$$\sum\sum x_{ij}^2 = 1162$$

1. Correction Factor: $CF = \frac{G^2}{n} = \frac{(116)^2}{15} = \frac{13456}{15} = 897.06$

2. Total Sum of Squares: $TSS = = \sum\sum x_{ij}^2 - C.F$

$= 1162 - 897.06 = 264.94$

3. Sum of Squares between Treatments: $SST = \frac{\sum x_i^2}{n_i} - C.F$

$$= \frac{5286}{5} - 897.06$$
$$= 1057.2 - 897.06$$
$$= 160.14$$

4. Sum of Squares due to Error: $SSE = TSS - SST$

$= 264.94 - 160.14 = 104.8$

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